

A Dual TOF Sensor System to Measure Room Capacity

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Team 2: PortalSense

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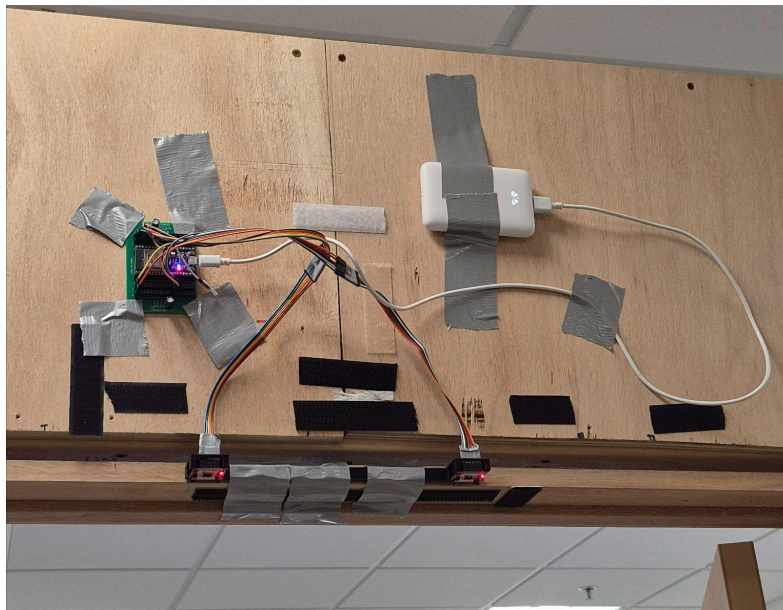


Figure 1: Door Mounted System during the Demo

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Introduction

This report presents the design and implementation of an Internet of Things (IoT) based sensor system for measuring the capacity of a room. The goal of this project was to develop a sensor suite that gathers real-time data on the number of people passing through a door and reports this data to a central server via a wireless communication channel. The error of this system should not exceed $\pm 5\%$ for a room that can hold up to 100 humans, and the system should be able to handle cases where 2 people pass through the door simultaneously, in either direction. The total R&D budget for this project was \$200 and the on-door cost per unit was constrained to \$100.

The final system is composed of two time-of-flight (TOF) sensors for data collection and an ESP32 microcontroller for data processing and transmission. A custom 3D printed mount was fabricated to mount the system to the top of a door frame and a custom printed circuit board (PCB) was designed and manufactured to simplify electrical connections (Figures 1 & 2). The system is battery-operated and transmits data using the Wi-Fi connectivity of the ESP32. The mass production of this system will result in a \$90.99 cost per unit.

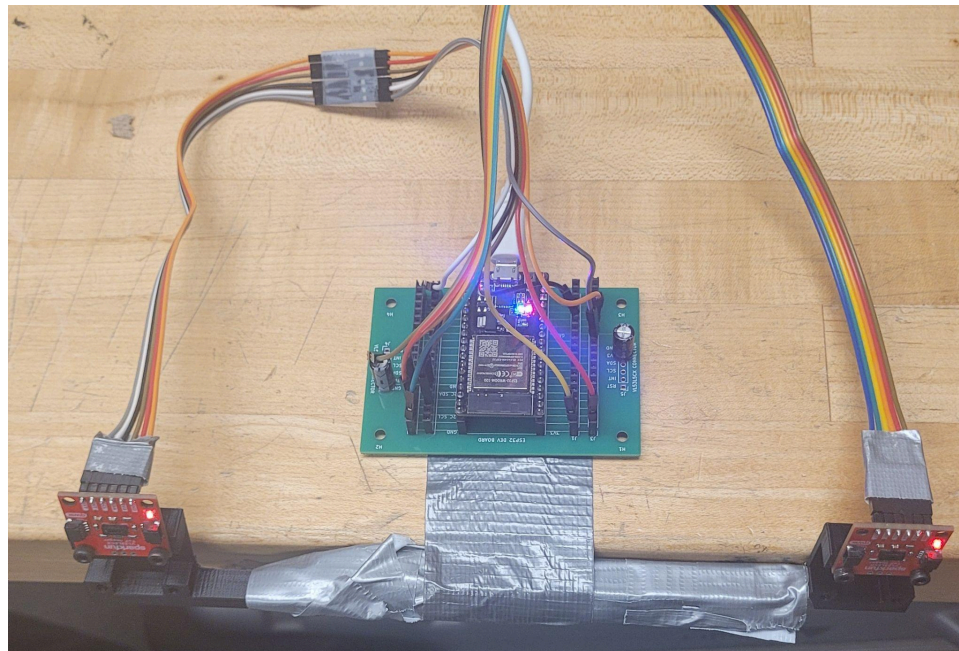


Figure 2: Fully Integrated System Hardware

Design Selection & Rationale

We evaluated several sensor combinations based on simplicity and estimated accuracy of the final system. We looked at three main options: (1) Dual IR sensors and Thermal Camera, (2) Time of Flight sensors, (3) Camera with Computer Vision Algorithm. Based on the long latency with the PIR sensors of option 1 and the software intensive nature of the computer vision algorithm of option 3, we decided on option 2 as it presented a balanced approach between long latency and very intensive software logic.

On the firmware side, the core algorithm was a finite state machine (FSM). This was chosen because it alleviated some initial timing concerns, as it locked the current state until a sensor or watchdog timer activated. The primary components of the FSM were pending states for entering, exiting, dual entering, dual exiting, along with idle and reset states. State transitions were primarily determined by sensor detections crossing certain thresholds. However, later in development, new timing concerns arose from said locking the state and the use of the watchdog timer and are further detailed in the Project Performance section.

Final Prototype

Overview

The final hardware design (figure 3) was composed of two VL53L5CX Time of Flight (TOF) sensors mounted on the left and right halves of the top of the door, along with the HiLetgo ESP32 breakout board and a battery pack on the center top of the door. TOF sensors were chosen for their advertised precision, being able to subdivide their FOV into 8x8 or 4x4 grids. The final design utilized the 4x4 grid mode as it offered a faster refresh rate and less edge distortion. For each sensor, the 16-cell grid was further subdivided into 2 main regions: outer and inner zones, which served as the enter/exit thresholds. These zones were further subdivided into zones for base case and side-by-side detection, netting 4 total subzones per sensor. Since only TOFs were used, the subdivision was necessary to tackle edge cases. This rationale is also why no other types of sensors were used. The detection distance threshold was 1.5m, which was necessary for detecting individuals in wheelchairs. This resulted in the outer zones covering the front of the door and the inner zones covering just passed the door's threshold, giving full coverage. To support these sensors and power, an ESP breakout PCB was included to support all the necessary electrical connections (figure 4 & 5).

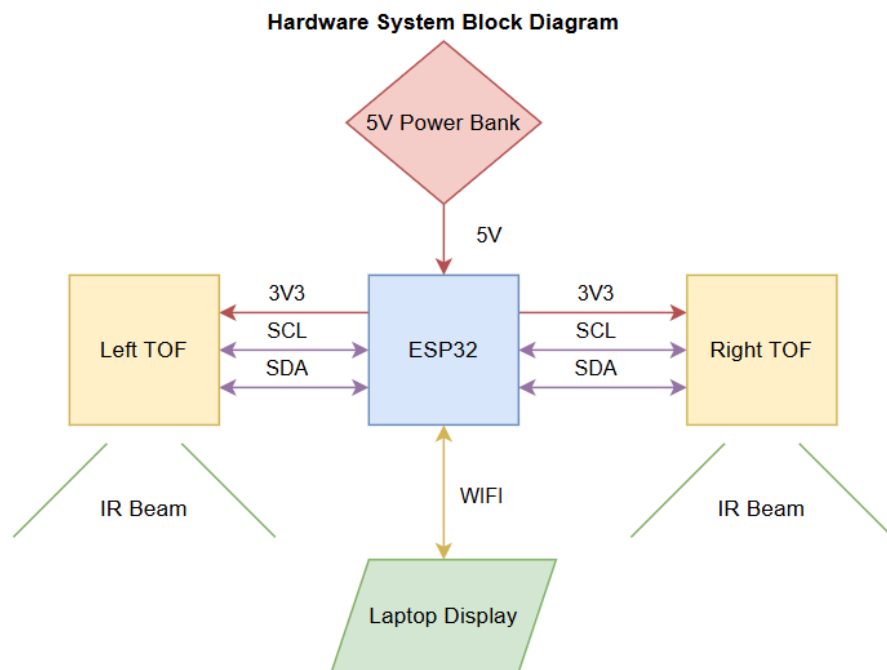


Figure 3: System Block Diagram.

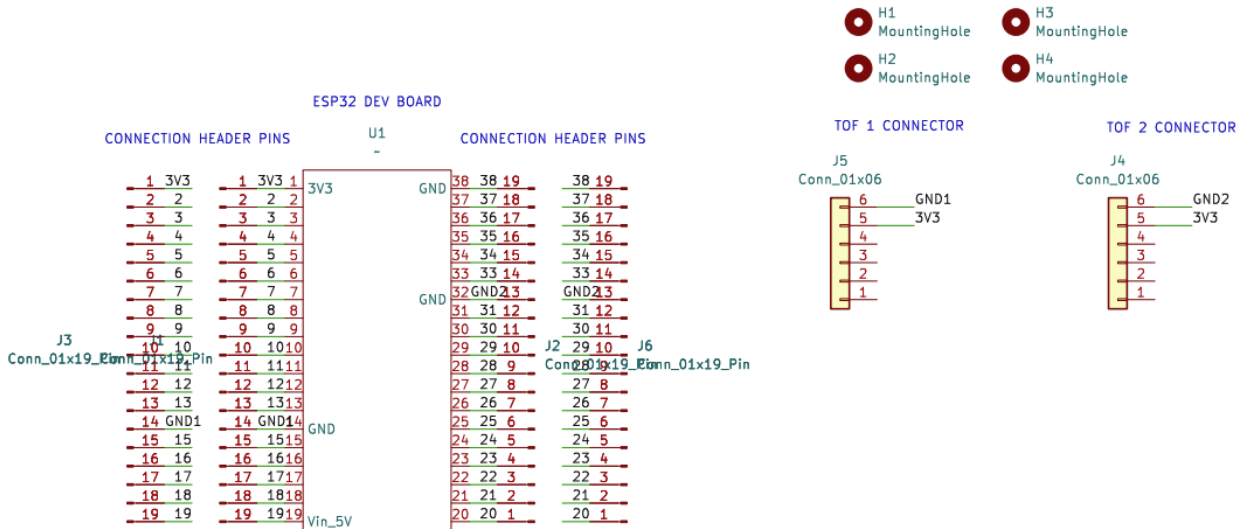


Figure 4: PCB Schematics

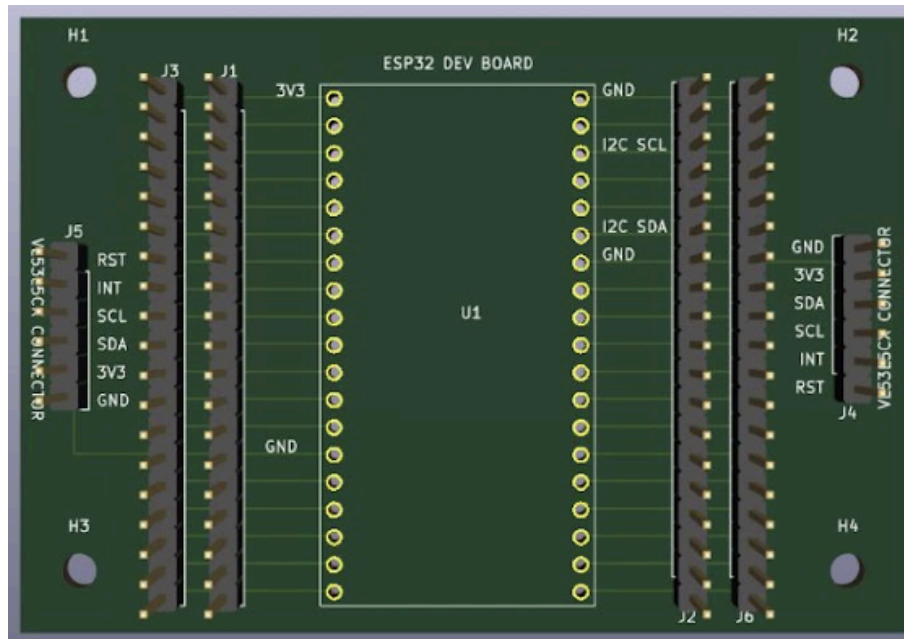


Figure 5: PCB Layout

Performance

The project performed moderately well on base case scenarios that included single file entry. For example the recorded congo line tests for entering and exiting had 85% and 90% accuracy for 20 people. The errors with these single person test cases can be attributed to timing errors. Take the entering condition, once the outer zone is triggered, the state transitions into an enter pending state where it awaits an inner zone trigger and ignores the outer zones until either the inner zone or the watchdog timer triggers. If during this time, a second person follows quickly, the device will not register that a second person has crossed the threshold, resulting in only a single person being counted. These timing issues extend to some of the other test cases such as: dual detection and partial entry. To account for mechanical misalignment and distortion on edges of the detection zones, multiple reading measurements are taken each

loop before running the FSM logic. However, these readings must happen fast enough to ensure that both outer and inner zones do not trigger simultaneously. Because of this, sometimes people are missed. Another feature of the VL53L5CX is that the ranging distance can be changed. Our device used a 1.5m ranging distance which is critical for the wheelchair case and enables reliable detection of a single person entering when closer to the ground. Lastly, the project remained under the \$100 on-door budget as seen in table 1.

<u>Item</u>	<u>Money Spent</u>	<u>Door Mass Production BOM</u>	<u>Net Off [\$200 Limit]</u>	<u>Net On [\$100 Limit]</u>
SparkFun VL53L5CX 8x8 TOF [2x]	X	\$65.00	\$36.40	\$90.99
ESP32 DevKit	X	\$10.00		
PCB [5x]	\$31.40	\$2.00		
Verical/Adafruit 5V 10Ah Battery Pack	X	\$8.99		
Screws & 3D Print Filament	X	\$5.00		
Surcharges	\$5.00	X		

Table 1: Project Expenditures & BoM

Power Analysis

The power analysis for the project was performed by measuring current and voltage via power supplies and inline resistors/oscilloscope voltage differentials. An average current consumption was measured to be around 240mA and there were brief 600mA spikes from WiFi and both TOFs sending ranging pulses simultaneously. The project utilized a 10Ah battery pack that did not supply a discharge curve. The 600mA spikes were extremely short, so the 250mA will be primarily used which yields a 40 hour battery life. However, because the internals of the pack is a 3v7 cell that boosts to 5V, there are some efficiency losses along with the fact that the pack will drop below the voltage threshold required to boost to 5V before 40 hours has passed. As such, a conservative estimate of 75% of the pack's usable capacity will be used. This brings the expected lifetime down to approximately 30 hours.

Future Improvements

One area that we could improve upon on future iterations is a more sophisticated mounting system. During testing and our final demo, we frequently had issues with the sensor position needing to be slightly angled or moved to improve our accuracy. Having a mount that spans the door would ensure that the sensors are always in the same position relative to the door and may solve this issue. Additional work to find a more optimal angle of the sensor to avoid the door frame intersecting the sensor's field of view would also improve the system's accuracy. The updated mount design would also include a mount for the PCB and battery to ensure the security of the system.

Theoretical Prototype Analysis

Our primary theoretical design alternative is a system combining Dual Infrared (IR) sensors with a low-resolution Thermal Camera. In this concept design, two pairs of IR emitters and receivers are mounted horizontally across the door frame at a specific height (lower third of the door) to act as “tripwires” (figure 6). A thermal camera (such as the Adafruit AMG8833 or MLX9064X) is mounted at the top of the door frame, facing downward.

This design solves the occupancy counting problem by separating the tasks of motion detection and object identification. The dual IR beams provide a robust, low-latency method for determining directionality (Entering vs. Exiting) based on the order in which the beams are broken. The thermal camera solves the problem of false positives from non-human objects. By analyzing the heat signature of the object crossing the threshold, the system can distinguish between a human and inanimate object (like a wheelchair).

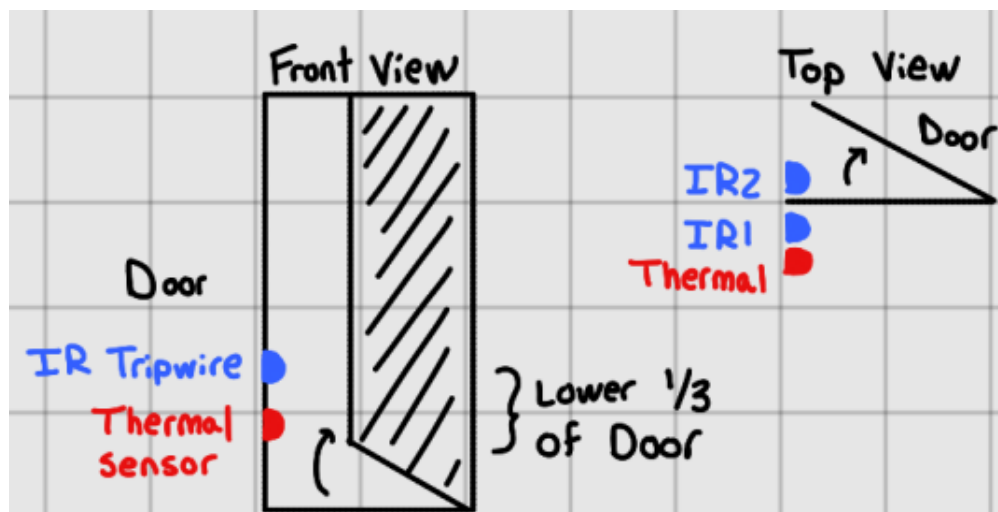


Figure 6: Theoretical System Layout. IR1 and IR2 create two distinct interrupt lines. The Thermal Sensor overlooks the transition zone.

This design relies on two cooperative algorithms running on the ESP32:

1. Directional Finite State Machine (FSM): similar to our final prototype, this monitors the binary state of the two IR receivers.
 - a. State ENTER_P: Triggered when Outer Beam (A) breaks while Inner Beam (B) is intact.
 - b. State EXIT_P: Triggered when Inner Beam (B) breaks while Outer Beam (A) is intact.
 - c. Valid transition requires the sequence $A \rightarrow A \& B \rightarrow B \rightarrow \text{Clear}$ (for entry).
2. Thermal Blob Analysis: The thermal camera provides a matrix of temperature readings
 - a. The algorithm applies a simple threshold filter. Any pixel $< 28^\circ\text{C}$ is zeroed out.
 - b. We could implement a Blob detection algorithm to identify distinct heat signatures.
 - c. A human count is registered only if the “heat blob” centroid moves across the IR beam center line.

Our team validated this design by analyzing component datasheets and project constraints. We examined the specifications for common thermal sensors, such as the AMG8833, and calculated that a standard 60° field-of-view sensor mounted at the door height of

2 meters would cover a floor area of roughly 2.3m x 2.3m. This analysis confirmed that the sensor could efficiently monitor the entire width of a standard doorway. However, there are some limitations to this design. Our financial analysis revealed a feasibility issue; an AMG8833 costs approximately \$40, combined with the necessary IR emitter/receiver pairs (\$50) and ESP32 (\$10), the sensor suite alone would consume the majority of the budget. This leaves insufficient funds for the custom PCB, battery, and mechanical enclosure. Additionally, we determined that while the IR logic is relatively simple, implementing the algorithm to process the thermal matrix data can combine it into a coherent FSM may take a significant amount of software knowledge and testing.

If a future team were to implement this design, we recommend taking steps to overcome the limitations we identified. Instead of the AMG8833 or MXX90640 thermal cameras, we would look for a lower cost, potentially lower resolution camera. To address the complexity inherent in thermal imaging we recommend having a group member with good software expertise and starting the blob detection algorithm early.

Conclusion

Through the development of PortalSense, our team concluded that Time-of-Flight (TOF) offers a superior balance of accuracy and privacy for indoor occupancy tracking compared to camera-based alternatives. While computer vision solutions may provide high-fidelity data, they raise privacy concerns. By utilizing the VL53L5CX's depth matrix, our system successfully anonymized the data at the hardware level while still maintaining necessary precision to detect entries. This project demonstrated that low-cost, battery-operated IoT devices can achieve reliable occupancy sensing without the high computational overhead or ethical risks associated with room surveillance. This makes this approach highly scalable for future pandemic or general applications.

Our final demonstration highlighted some limitations of our software logic. Our decision to static zone triggering resulted in consistent base case but failures in certain edge-case scenarios like bi-directional traffic and tailgating. In these instances, the static zones would merge the two individuals into a single continuous active signal, causing the FSM to register only one count instead of two. Similarly, side-by-side detection was inconsistent; because we relied on specific small zones to detect width, slight variations in how people walked through the doorframe often led to missed detections or false positives.

Our advice to future students is to look beyond simple thresholding for any application involving multiple targets. Future teams should consider implementing centroid tracking or Blob Detection algorithms. While we proved that simple thresholding can be effective, these methods allow the system to identify distinct entities and track their unique paths, which is the most effective way to disentangle close-proximity targets like tailgaters or side-by-side walkers. This project has taught us a lot about the software development process. We recommend building a software simulator for your ESP logic early in the design process. Simulating sensor data on a laptop allows you to validate your counting logic, saving valuable time for hardware integration later on.

Appendix

Included in the “[Final paper team 2 supporting files](#)” folder are the following:

KiCad PCB Files:

- PCB was developed in KiCad
- “EECS300.pretty” contains custom footprint for ESP32 dev board

CAD Files:

- Prt file: [sensorMountClean.prt](#)
- Drawing PDF:  eeecs300SensorMount.pdf

Firmware Files:

- main.ino → Contain all code logic like the counter, WIFI, FSM, etc.
- SparkFun_VL53LCX_Arduino_Library → TOF interfacing library for the ESP that allows for easier data readings
- i2cdetect → Debugging library to scan an I2C bus to detect if devices are detectable and report their addresses

README.txt